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EXAMINER	
NGUYEN, TRAN N	

ART UNIT	PAPER NUMBER
3685	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/438,563	Applicant(s) KIM et al.	
	Examiner TRAN NGUYEN	Art Unit 3685	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 30 December 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,4-5,7-11 and 15 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,4-5,7-11 and 15 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claim(s) 1,4-5,7-11 and 15 is/are rejected under 35 U.S.C. 101 because the claimed invention is directed to abstract idea without significantly more.

Claim 1 recites:

An apparatus for predicting hypotension of a subject, comprising:

a memory configured to store one or more instructions and a pre-trained hypotension prediction model; and

a processor configured to execute the one or more instructions stored in the memory, wherein the instructions, when executed by the processor, cause the processor to:

determine an arterial blood pressure data of the subject,

input the arterial blood pressure data of the subject into the hypotension prediction model, and

determine whether the subject has hypotension using an output result of the hypotension prediction model,

wherein the hypotension prediction model is trained to determine whether a training subject has hypotension for a training arterial blood pressure data of the training subject,

wherein the hypotension prediction model is trained by a training input data including a plurality of intervals of the training arterial blood pressure data of the training subject and a label data including whether or not hypotension has occurred for each of the plurality of intervals of the training arterial blood pressure data,

wherein the hypotension prediction model includes:

a first layer **trained to extract trend data for each of the plurality of intervals of the training arterial blood pressure data of the training subject by wavelet-transforming the plurality of intervals of the training arterial blood pressure data of the training subject**; and

a shapelet data generation module configured to **generate a shapelet data corresponding to the trend data**,

wherein the shapelet data represents a feature for a hypotension prediction corresponding to a local shape of a predetermined interval within time-series data, and

wherein the processor is configured to input the arterial blood pressure data of the subject to the hypotension prediction model, and **to calculate similarity between trend data for each of the plurality of intervals of the arterial blood pressure data of the subject and the shapelet data generated** by the shapelet data generation module.

Step 1:

The claim as a whole falls within at least one statutory category, i.e. a process, machine, manufacture, or composition of matter.

Step 2A Prong One:

The highlighted portion, as drafted, is a process that, under its broadest reasonable interpretation, falls under "Mathematical concepts" because the steps of training a model and extracting trend data are directed towards mathematical calculations, i.e. mathematical relationships, mathematical formulas or equations, mathematical calculations. MPEP § 2106.04(a)(2)(I)

The highlighted portion, as drafted, is a process that, under its broadest reasonable interpretation, falls under "Certain methods of organizing human activity" because the steps making determination and using a trained model are typically performed by human beings, i.e. fundamental economic principles or practices (including hedging, insurance, mitigating risk); commercial or legal interactions (including agreements in the form of contracts; legal obligations; advertising, marketing or sales activities or behaviors; business relations); managing personal behavior or relationships or interactions between people (including social activities, teaching, and following rules or instructions). MPEP 2106.04(a)(2)(II)

The highlighted portion, as drafted, is a process that, under its broadest reasonable interpretation, falls under "Mental processes".

But for a generic computer invoked with a high level of generality in a *post hoc* manner to implement the abstract idea, the determining steps may be performed in the human mind either mentally or with pen and paper.

Accordingly, these limitations have been found to be directed towards concepts performed in the human mind (including an observation, evaluation, judgment, opinion). MPEP 2106.04(a)(2)(III)

The different categories of abstract ideas are being considered together as one single abstract idea. MPEP 2106.04(II)(B)

Dependent claim(s) recite(s) additional subject matter which further narrows or defines the abstract idea embodied in the claims (such as claim(s) 4-5, 7 reciting limitations further defining the abstract idea, which may be performed in the mind but for recitation of generic computer components, and/or may be a method of managing relationship or interactions between people, including various mathematical concepts such as regression and weighted trendline analysis).

Step 2A Prong Two:

This judicial exception is not integrated into a practical application. In particular, the claim recites the following additional element(s), if any:

a memory configured to store one or more instructions and a pre-trained hypotension prediction model; and

a processor configured to execute the one or more instructions stored in the memory, wherein the instructions, when executed by the processor, cause the processor to:

input the arterial blood pressure data of the subject into the hypotension prediction model,

a first layer; and

a shapelet data generation module;

wherein the processor is configured to input the arterial blood pressure data of the subject to the hypotension prediction model.

The additional element(s) do(es) not integrate the abstract idea into a practical application, other than the abstract idea *per se*.

Regarding the memory, the Specification as originally filed on 12 February 2024 discloses generic types of memories (page 9 paragraph 046).

Regarding the processor, the Specification discloses a general-purpose computer (page 18-19 paragraph 127).

Accordingly, these limitations amount(s) to mere instructions to apply an exception (invoking computers as a tool to perform the abstract idea). MPEP 2106.05(f))

Regarding the step of inputting data and the layers of the prediction model, these limitations merely add(s) insignificant extra-solution activity to the abstract idea. MPEP 2106.05(g))

Dependent claim(s) recite(s) additional subject matter which amount to limitation(s) consistent with the additional element(s) in the independent claims (such as claim(s) 5 reciting layers of the prediction model, additional limitation(s) which add(s) insignificant extra-solution activity to the abstract idea by selecting a particular data source or type of data to be manipulated).

Looking at the limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually. There is no indication that the combination of elements improves the functioning of a computer or improves any other technology. Their collective functions merely provide conventional computer implementation and do not impose a meaningful limit to integrate the abstract idea into a practical application.

Accordingly, the additional elements do not integrate the judicial exception into a practical application because it does not impose any meaningful limits on practicing the abstract idea.

Accordingly, the claim recites an abstract idea.

Step 2B:

The claim does not include additional elements that are sufficient to amount to significantly more than the judicial exception. As discussed above with respect to integration of the abstract idea into a practical application, the additional elements amount to no more than mere instructions to apply an exception, add insignificant extra-solution activity to the abstract idea, and/or generally link the abstract idea to a particular technological environment or field of use.

The additional elements, as discussed above and incorporated herein, amount to no more than mere instructions to apply an exception, add insignificant extra-solution activity to the abstract idea, and/or generally link the abstract idea to a particular technological environment or field of use, as discussed above and incorporated herein.

Mere instructions to apply an exception, insignificant extra-solution activity, and linking to a particular technological environment using a generic computer component cannot provide an inventive concept.

Regarding the step of inputting data into the prediction model, this limitation amount(s) to element(s) that have been recognized as well-understood, routine, and conventional (WURC) activity in particular fields (e.g., receiving or transmitting data over a network, Symantec, MPEP 2106.05(d)(II)(i), e.g., storing and retrieving information in memory, Versata Dev. Group, MPEP 2106.05(d)(II)(iv)). MPEP 2106.05(d)(II)(ii))

Regarding the layers of the prediction model, Yabuuchi (20210257067) discloses a multi-layer neural network used for a prediction model in a manner that would be WURC (page 5 paragraph 0064, e.g. “for example” language).

Dependent claims recite additional subject matter which amount to limitations consistent with the additional elements in the independent claims (such as claim(s) 5 reciting layers of the

prediction model; Yabuuchi discloses a multi-layer neural network used for a prediction model in a manner that would be WURC (page 5 paragraph 0064, e.g. "for example" language)). MPEP 2106.05(d)(II)(ii))

Looking at the limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually. There is no indication that the combination of elements improves the functioning of a computer or improves any other technology. Their collective functions merely provide conventional computer implementation.

The claim is not patent eligible.

Claim(s) 8, 9, 10, 11, 15 recite(s) substantially similar limitations as those of claim(s) 1, 1, 5, 1, 7 above, and are therefore rejected for substantially similar rationale as applied above, and incorporated herein.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claim(s) 1, 4-5, 7-11, 15 is/are rejected under 35 U.S.C. 103 as being unpatentable over Olde (20120283581) in view of Angelotti (Prediction and characterization of acute hypotensive episodes in intensive care unit, previously mailed on 30 September 2025).

Claim 1: Olde discloses:

An apparatus (page 5 paragraph 0067 illustrating an apparatus) for predicting hypotension of a subject (page 8-9 paragraph 0130 illustrating predicting hypotension for a patient [considered to be a form of "subject"]), comprising:

a memory configured to store one or more instructions (page 11-12 paragraph 0165 illustrating a memory storing instructions thereon) and a pre-trained hypotension prediction model (page 33 paragraph 0509 illustrating a model with account arrangement-specific parameters [considered to be a form of "pre-trained"]); and

a processor configured to execute the one or more instructions stored in the memory, wherein the instructions (page 11-12 paragraph 0165 illustrating a processor), when executed by the processor, cause the processor to:

determine an arterial blood pressure data of the subject (Figure 8 illustrating measuring blood pressure for the patient),

input the arterial blood pressure data of the subject into the hypotension prediction model (page 35-36 paragraph 0531 illustrating entering blood pressure data into the model), and

determine whether the subject has hypotension using an output result of the hypotension prediction model (page 10 paragraph 0153 illustrating classifying the patient regarding whether they will have hypertension),

wherein the hypotension prediction model is trained to determine whether a training subject has hypotension for a training arterial blood pressure data of the training subject (page 23 paragraph 0370 illustrating using a bandpass filter to process blood pressure data within a particular acceptable range [considered to be a form of "training" data]),

wherein the hypotension prediction model is trained by a training input data including a plurality of intervals of the training arterial blood pressure data of the training subject (page 36 paragraph 0556 illustrating using a plurality of frequency intervals in the .5-2.7 Hz range to indicate data of interest for use to further process the data [considered to be a form of "training"]) and a label data including whether or not hypotension has occurred for each of the plurality of intervals of the training arterial blood pressure data (page 10 paragraph 0153 illustrating determining specific criteria for hypotension present in patients with dialysis),

wherein the hypotension prediction model includes:

~~a first layer trained to extract trend data for each of the plurality of intervals of the training arterial blood pressure data of the training subject by wavelet-transforming the plurality of intervals of the training arterial blood pressure data of the training subject~~; and

a shapelet data generation module configured to generate a shapelet data corresponding to the trend data (page 5 paragraph 0063 illustrating determining trending of the blood pressure shape data),

wherein the shapelet data represents a feature for a hypotension prediction corresponding to a local shape of a predetermined interval within time-series data (Figure 2 illustrating a prediction that involves a time series on the x-axis), and

wherein the processor is configured to input the arterial blood pressure data of the subject to the hypotension prediction model (as discussed above and incorporated herein), and ~~to calculate similarity between trend data for each of the plurality of intervals of the arterial blood pressure data of the subject and the shapelet data generated by the shapelet data generation module~~.

Olde does not disclose:

a first layer trained to extract trend data for each of the plurality of intervals of the training arterial blood pressure data of the training subject by wavelet-transforming the plurality of intervals of the training arterial blood pressure data of the training subject; and

to calculate similarity between trend data for each of the plurality of intervals of the arterial blood pressure data of the subject and the shapelet data generated by the shapelet data generation module.

Angelotti discloses:

a first layer trained to extract trend data for each of the plurality of intervals of the training arterial blood pressure data of the training subject by wavelet-transforming the plurality of intervals of the training arterial blood pressure data of the training subject (page 15 paragraph 2 illustrating a multi-layer neural network used to calculate wavelet coefficients for blood pressure, page 9 paragraph 2 illustrating 20 second intervals); and

to calculate similarity between trend data for each of the plurality of intervals of the arterial blood pressure data of the subject and the shapelet data generated by the shapelet data

generation module (page 76 last paragraph illustrating measuring the Euclidean distance to measure for similarity between data points).

Before the effective filing date of the claimed invention, it would have been obvious to one of ordinary skill in the art to include the machine learning techniques of Angelotti within the blood pressure management system of Olde with the motivation of improving patient care in intensive care units by identifying and quickly treating hypotension (Angelotti; page xi Abstract).

Claim 4: Olde in view of Angelotti disclose:

The apparatus of claim 1, as discussed above and incorporated herein.

Olde does not disclose:

wherein the hypotension prediction model is trained to low-pass filter the training arterial blood pressure data using training parameters trained in the first layer.

Angelotti discloses:

wherein the hypotension prediction model is trained to low-pass filter the training arterial blood pressure data using training parameters trained in the first layer (page 24 last paragraph illustrating a third order lowpass filter used to filter blood pressure data).

Before the effective filing date of the claimed invention, it would have been obvious to one of ordinary skill in the art to include the machine learning techniques of Angelotti within the blood pressure management system of Olde in view of Angelotti with the motivation of improving patient care in intensive care units by identifying and quickly treating hypotension (Angelottie; page xi Abstract).

Claim 5: Olde in view of Angelotti disclose:

The apparatus of claim 1, as discussed above and incorporated herein.

Olde does not disclose:

wherein the hypotension prediction model includes:

a second layer trained to assign a weight to at least one of intervals among the plurality of intervals of the training arterial blood pressure data of the training subject; and

a third layer trained to calculate similarity feature value between shapelet data of the at least one interval and the trend data on the basis of the assigned weight.

Angelotti discloses:

a second layer trained to assign a weight to at least one of intervals among the plurality of intervals of the training arterial blood pressure data of the training subject (page 50 paragraph 2 illustrating weighting the series); and

a third layer trained to calculate similarity feature value between shapelet data of the at least one interval and the trend data on the basis of the assigned weight (page 76 last paragraph illustrating measuring the Euclidean distance to measure for similarity between data points).

Before the effective filing date of the claimed invention, it would have been obvious to one of ordinary skill in the art to include the machine learning techniques of Angelotti within the blood pressure management system of Olde in view of Angelotti with the motivation of improving patient care in intensive care units by identifying and quickly treating hypotension (Angelotti; page xi Abstract).

Claim 7: Olde in view of Angelotti disclose:

The apparatus of claim 1, as discussed above and incorporated herein.

Olde does not disclose:

wherein the processor is configured to calculate hypotension probability for the calculated similarity of the trend data for each of the plurality of intervals of the arterial blood pressure data of the subject using a logistic regression layer.

Angelotti discloses:

wherein the processor is configured to calculate hypotension probability for the calculated similarity of the trend data (page 3 paragraph 2 illustrating probability models) for each of the plurality of intervals of the arterial blood pressure data of the subject using a logistic regression layer (page xr illustrating a regression logistics).

Before the effective filing date of the claimed invention, it would have been obvious to one of ordinary skill in the art to include the machine learning techniques of Angelotti within the blood pressure management system of Olde in view of Angelotti with the motivation of improving patient care in intensive care units by identifying and quickly treating hypotension (Angelotti; page xi Abstract).

Claim(s) 8, 9, 10, 11, 15 recite(s) substantially similar limitations as those of claim(s) 1, 1, 5, 1, 7 above, and are therefore rejected for substantially similar rationale as applied above, and incorporated herein.

Response to Arguments

In the Remarks filed on 30 December 2025, Applicant makes numerous arguments. Examiner will address these arguments in the order presented.

On page 7-8 Applicant argues that the Office did not perform a proper analysis under the Alice/Mayo two-part framework, and argues specifically on page 8 that the Office did not examine the claims as a whole nor in light of the Specification.

In making this argument, Applicant provides no specific evidence that the claims were not examined as a whole nor in light of the Specification.

To the contrary, as in the section above and incorporated herein, the Office action has meticulously highlighted portions of the claim(s) considered to be directed towards an abstract idea, to include reasoned rationale regarding which enumerated categories of abstract idea to which these limitations belong.

Furthermore, the claims as a whole were considered in light of the Specification as understood by one of ordinary skill in the art.

Accordingly, Applicant's argument amounts to mere general allegation of patentability, without specifically pointing out how the claims were not considered as a whole, nor in light of the Specification.

On page 8-10 Applicant argues that the claims provide a specific way of predicting if a subject has/will have hypotension using a specific method, and involves a specific hypotension prediction model that uses specific types of blood pressure data analyzed in specific ways.

As cited by Applicant, Synopsys provides eligibility for specific data encryption method, and accordingly, the Office has long recognized encryption to be patent eligible subject matter as being a recognized technology. Consistently, the Office recognizes that encryption algorithms are typically not considered part of Mental Processes because they cannot be practically performed in the human mind.

In the instant pending claims, there is no encryption recited, and the various data processing steps may be practically performed in the human mind, for the reasons stated above and incorporated herein.

On page 10-12, Applicant argues that the claims improve another technology or technical field, such as human healthcare and/or other computer hardware associated with surgical monitoring.

Examiner maintains that the claims do not improve any technical field because to the extent that there are any improvement, they are limited to the abstract idea highlighted in the section above, and incorporated herein.

Even newly discovered or novel judicial exceptions are still exceptions. MPEP 2106.04(I).

It is noted that in making this argument, Applicant does not specifically state which additional element(s) would provide such technical improvement.

On page 13-14 Applicant argues that the applied art do not fairly disclose or suggest an analysis involving shapelets, and specifically shapelets that correspond to distinct sub-intervals in the times-series data and describe specific features (local shapes) in that time-series data.

Examiner contends that, as described in the section above, that the argued features (to the extent that they are claimed) are fully disclosed and suggested by the applied art.

For example, Applicant argues that the applied art do not disclose comparison of time-series data to shapelets, generation of shapelets, or decomposition of data into shapelets;

however, on page 13 Applicant asserts that Angelotti discloses a method that involves wavelet decomposition and weighting thereof to analyze a signal.

The claimed features are fully disclosed in the combined applied art, as discussed above and incorporated herein.

Based on the evidence presented above, Applicant's arguments are not found persuasive.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Craine (20120290318) discloses tracking a claim to detect appropriate treatment and diagnosis of a patient (Abstract) in a manner similar to those disclosed in the instant pending Specification as originally filed.

Allen (20180096103) discloses analyzing medical records using machine learning to verify a diagnosis for a patient (Abstract) in a manner similar to those disclosed in the instant pending Specification as originally filed.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after

the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to TRAN N NGUYEN whose telephone number is (571)272-0259. The examiner can normally be reached Monday-Friday 9AM-5PM Eastern.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, KAMBIZ ABDI can be reached on (571)272-6702. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service

Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/T.N.N./
Examiner, Art Unit 3685

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